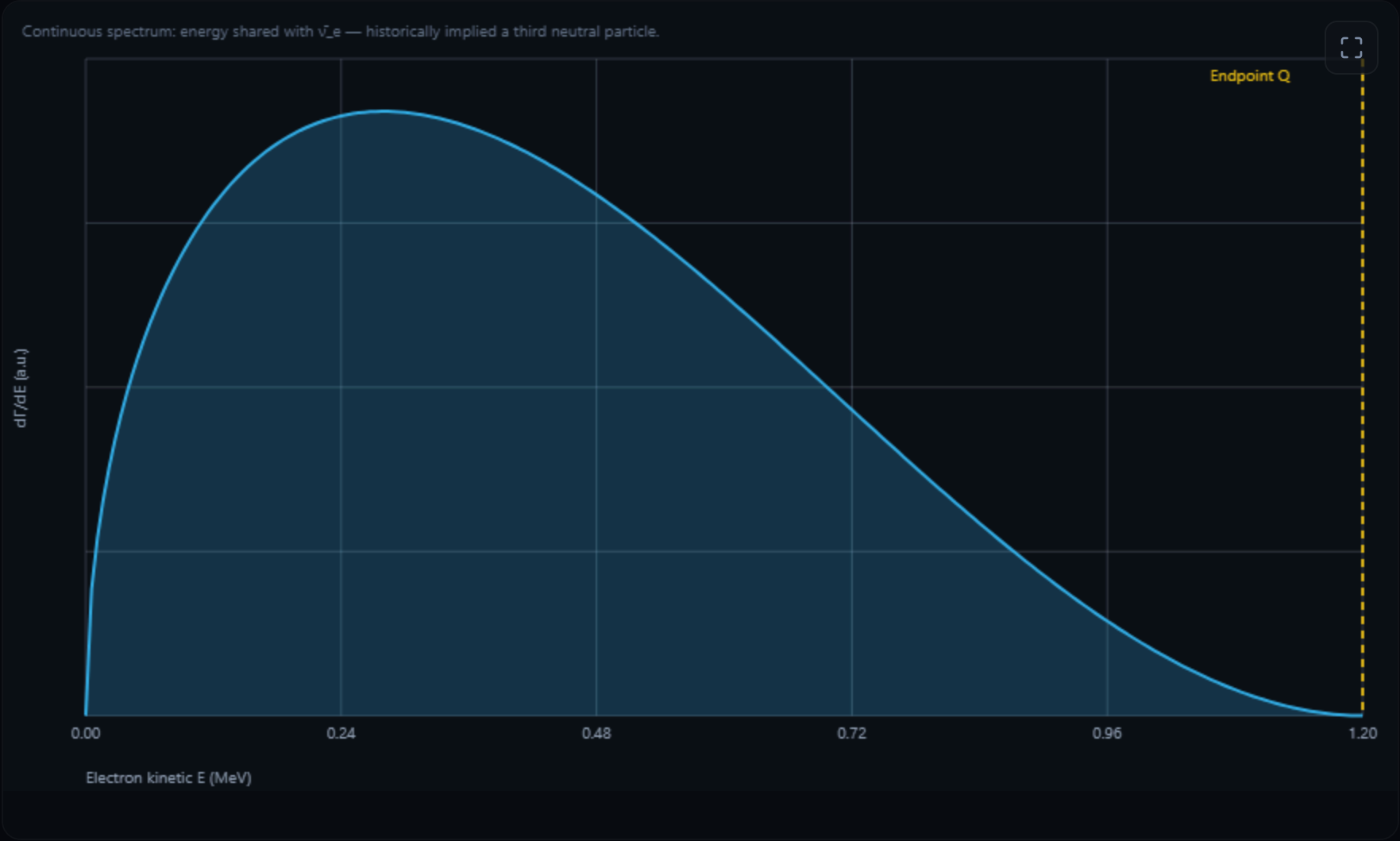


# $\beta^-$ Decay: Electron Spectrum

Continuous kinetic-energy spectrum up to endpoint Q (allowed-decay phase space); missing energy carried by the antineutrino.



## ALLOWED $\beta^-$ SPECTRUM

Endpoint Q (max electron T) 1.2 MeV

Toy Fermi phase space for an allowed decay:  
 $d\Gamma/dE \propto p (Q-E)^2$  with relativistic  $p = \sqrt{E^2 + 2 m_e E}$ . Coulomb correction  $F(Z,E)$  is omitted. The yellow line marks the endpoint  $E = Q$ .

## MEASURED VALUES

|                             |            |
|-----------------------------|------------|
| Q (endpoint)                | 1.200 MeV  |
| $\langle E \rangle$ (shape) | 0.4204 MeV |

### About this model

In  $\beta^-$  decay a nucleus emits an electron and an electron antineutrino, sharing energy and momentum with the recoiling daughter. For an allowed transition the electron kinetic-energy spectrum is continuous from zero up to an endpoint Q (the maximum electron kinetic energy when the antineutrino carries essentially none). A standard Fermi “golden rule” phase-space cartoon gives  $d\Gamma/dE \propto p (Q - E)^2$  with relativistic electron momentum  $p(E) = \sqrt{E^2 + 2 m_e c^2 E}$ . Coulomb distortion  $F(Z,E)$  near the nucleus is omitted here, so the curve is schematic but shows the key qualitative shape: rising from threshold, peaking, then falling to zero at  $E = Q$ . The mean kinetic energy  $\langle E \rangle$  for the displayed shape is shown for comparison when you change Q.

**Who it's for:** Modern physics and introductory nuclear physics courses discussing neutrinos, lepton number, and kinematics of three-body decay.

KEY TERMS

- Beta Decay
- Fermi Golden Rule
- Endpoint Energy
- Antineutrino
- Continuous Spectrum

How it works

Electron kinetic-energy spectrum in  $\beta^-$  decay is continuous up to an endpoint  $Q$ : the antineutrino carries the missing energy and momentum — Pauli’s 1930 hypothesis, confirmed by experiment.

Frequently asked questions

Why is the spectrum not symmetric around  $Q/2$ ?

The  $(Q - E)^2$  factor favors lower electron energies because more phase space opens when the light antineutrino takes more energy; the electron momentum factor  $p(E)$  also shapes the low-energy rise. Real nuclei add a Coulomb correction that modifies the low-energy end.

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